KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6492A | 6 Credits: 4 | Open Elective | Source-grounded

Introduction to IoT

- Provide basic concepts of Internet of Things.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6492A is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6492A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Microelectronics
Course code	6492A
Course title	Introduction to IoT
Semester	6 Credits: 4
Course category	Open Elective
Periods per week	4 (L:3 T:1 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● Provide basic concepts of Internet of Things.

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding Things (IoT) Illustrate the IoT applications using Embedded	See official syllabus
CO2	14 Applying Computing Board - Arduino Identify the Python programming constructs for	See official syllabus
CO3	12 Applying Illustrate the IoT applications using Embedded	See official syllabus
CO4	20 Understanding Computing Board - Raspberry PI	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3
CO3	CO3 3
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the fundamental concepts of Internet of Things (IoT)	5	As specified
Module 2	Arduino	4	As specified
Module 3	Identify the Python programming constructs for IoT	4	As specified
Module 4	Raspberry PI	3	As specified

MODULE 1

Explain the fundamental concepts of Internet of Things (IoT)

This module is presented from the official syllabus scope for Introduction to IoT. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the fundamental concepts of Internet of Things (IoT)
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

2 Understanding. (IoT). Interpret IoT hardware, protocols and

2 Understanding. (IoT). Interpret IoT hardware, protocols and is an official topic in Module 1 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. (IoT). Interpret IoT hardware, protocols and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. (iot). interpret iot hardware, protocols and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. (IoT). Interpret IoT hardware, protocols and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] 2 Understanding. (IoT). Interpret IoT hardware, protocols and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding. software using IoT Stack. Summarize different IoT enabling

3 Understanding. software using IoT Stack. Summarize different IoT enabling is an official topic in Module 1 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding. software using IoT Stack. Summarize different IoT enabling using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding. software using iot stack. summarize different iot enabling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding. software using IoT Stack. Summarize different IoT enabling means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ 3 Understanding. software using IoT Stack. Summarize different IoT enabling □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Understanding. technologies. Classify IoT based on the complexity to build

3 Understanding. technologies. Classify IoT based on the complexity to build is an official topic in Module 1 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding. technologies. Classify IoT based on the complexity to build using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding. technologies. classify iot based on the complexity to build should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding. technologies. Classify IoT based on the complexity to build means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] 3 Understanding. technologies. Classify IoT based on the complexity to build [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding. and operate.

3 Understanding. and operate. is an official topic in Module 1 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding. and operate. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding. and operate. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding. and operate. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 3 Understanding. and operate. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain challenges in IoT 1 Understanding Introduction and Definition of IoT, Applications – Characteristics - Things in IoT - IoT Stack – IoT enabling Technologies – IoT challenges – IoT Levels Illustrate the IoT applications using Embedded Computing Board -

Explain challenges in IoT 1 Understanding Introduction and Definition of IoT, Applications – Characteristics - Things in IoT - IoT Stack – IoT enabling Technologies – IoT challenges – IoT Levels Illustrate the IoT applications using Embedded Computing Board - is an official topic in Module 1 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain challenges in IoT 1 Understanding Introduction and Definition of IoT, Applications – Characteristics - Things in IoT - IoT Stack – IoT enabling Technologies – IoT challenges – IoT Levels Illustrate the IoT applications using Embedded Computing Board - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain challenges in iot 1 understanding introduction and definition of iot, applications – characteristics - things in iot - iot stack – iot enabling technologies – iot challenges – iot levels illustrate the iot applications using embedded computing board - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain challenges in IoT 1 Understanding Introduction and Definition of IoT, Applications – Characteristics - Things in IoT - IoT Stack – IoT enabling Technologies – IoT challenges – IoT Levels

Aspect	What to prepare
	Illustrate the IoT applications using Embedded Computing Board - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain challenges in IoT 1 Understanding Introduction and Definition of IoT, Applications – Characteristics - Things in IoT - IoT Stack – IoT enabling Technologies – IoT challenges – IoT Levels Illustrate the IoT applications using Embedded Computing Board - □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the fundamental concepts of Internet of Things (IoT)

              Outline the key features of Internet of Things
   M1.01                                                      2
Understanding.
              (IoT).
              Interpret IoT hardware, protocols and
   M1.02                                                      3
Understanding.
              software using IoT Stack.
              Summarize different IoT enabling
   M1.03                                                      3
Understanding.
              technologies.
              Classify IoT based on the complexity to build
   M1.04                                                      3
Understanding.
              and operate.
   M1.05      Explain challenges in IoT                       1
Understanding
 Contents:
 Introduction and Definition of IoT, Applications – Characteristics - Things in IoT
- IoT
 Stack – IoT enabling Technologies – IoT challenges – IoT Levels

              Illustrate the IoT applications using Embedded Computing Board -
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Arduino

This module is presented from the official syllabus scope for Introduction to IoT. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Arduino
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing

Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing is an official topic in Module 2 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the working of sensors and actuators 2 understanding outline the features of embedded computing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Understanding boards - Arduino. Illustrate the interfacing of sensors and

3 Understanding boards - Arduino. Illustrate the interfacing of sensors and is an official topic in Module 2 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding boards - Arduino. Illustrate the interfacing of sensors and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding boards - arduino. illustrate the interfacing of sensors and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding boards - Arduino. Illustrate the interfacing of sensors and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ 3 Understanding boards - Arduino. Illustrate the interfacing of sensors and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

5 Applying actuators with Arduino Illustrate the interaction with web page and

5 Applying actuators with Arduino Illustrate the interaction with web page and is an official topic in Module 2 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying actuators with Arduino Illustrate the interaction with web page and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying actuators with arduino illustrate the interaction with web page and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Applying actuators with Arduino Illustrate the interaction with web page and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 5 Applying actuators with Arduino Illustrate the interaction with web page and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud.

4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud. is an official topic in Module 2 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying cloud sensors and actuators - role of sensors and actuators in iot - working of sensors and actuators - examples. embedded computing boards - features and characteristics of arduino/node mcu, interfacing with arudino, configure arduino for iot, transferring data from web and cloud. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Arduino
   M2.01      Illustrate the working of sensors and actuators       2
Understanding
              Outline the features of embedded computing
   M2.02                                                            3
Understanding
              boards - Arduino.
              Illustrate the interfacing of sensors and
   M2.03                                                            5
Applying
              actuators with Arduino
              Illustrate the interaction with web page and
  M 2.04                                                            4
Applying
              cloud
              Series Test I                                         1
Contents:
Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors
and
Actuators - Examples.
Embedded Computing Boards - features and characteristics of Arduino/Node MCU,
Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and
Cloud.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Identify the Python programming constructs for IoT

This module is presented from the official syllabus scope for Introduction to IoT. Use the topic cards for learning and the source transcript for exact coverage.

As specified **CO-linked**

official hours | see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the Python programming constructs for IoT
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding. Python Illustrate the working of control structures in

2 Understanding. Python Illustrate the working of control structures in is an official topic in Module 3 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. Python Illustrate the working of control structures in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. python illustrate the working of control structures in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. Python Illustrate the working of control structures in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 2 Understanding. Python Illustrate the working of control structures in □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Applying. Python Develop python programs to solve simple

4 Applying. Python Develop python programs to solve simple is an official topic in Module 3 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying. Python Develop python programs to solve simple using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying. python develop python programs to solve simple should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying. Python Develop python programs to solve simple means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 4 Applying. Python Develop python programs to solve simple □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Applying. applications

4 Applying. applications is an official topic in Module 3 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying. applications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying. applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying. applications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Applying. applications □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board -

Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board - is an official topic in Module 3 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, make use of python packages for iot 2 applying. python - variables and data types - executing a python program - control structures - python module - simple programs - python for iot. illustrate the iot applications using embedded computing board - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for

Aspect	What to prepare
	IoT. Illustrate the IoT applications using Embedded Computing Board - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board - □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Identify the Python programming constructs for IoT

             Explain the basic programming concepts of
  M3.01                                                           2
Understanding.
             Python
             Illustrate the working of control structures in
  M3.02                                                           4
Applying.
             Python
             Develop python programs to solve simple
  M3.03                                                           4
Applying.
             applications
  M3.04      Make use of Python packages for IoT                  2
Applying.

Contents:
Python - variables and data types - executing a python program - control structures
-
python module - simple programs - python for IoT.

             Illustrate the IoT applications using Embedded Computing Board -
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Raspberry PI

This module is presented from the official syllabus scope for Introduction to IoT. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Raspberry PI
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with

2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with is an official topic in Module 4 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding boards - raspberry pi illustrate the interfacing of basic sensors with should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart

8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart is an official topic in Module 4 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding embedded computing board - raspberry pi case study - the applications building with iot - smart perishable tracking/smart transportation, smart healthcare, smart should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/

10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/ is an official topic in Module 4 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 10 understanding lavatory maintenance, smart water through iot, smart warehouse monitoring, smart retail, smart driver assistance system etc programming raspberry pi with python, interfacing sensors and actuators with raspberry pi. case study - applications building with iot- smart perishable tracking/smart transportation, smart healthcare, smart lavatory maintenance, smart water through iot, smart warehouse monitoring, smart retail, smart driver assistance system. text / reference t/r book title/author rmd sundaram shriram k vasudevan, abhishek s nagarajan, internet of t1 things, wiley publications vijay madisetti, arshdeep bahga, internet of things: a hands-on approach, r1 orient blackswan raj kamal, internet of things : architecture and design principles, mcgraw r2 hill education r3 gary smart, practical python programming for iot, packt publishing r4 marco schwartz, internet of things with arduino cookbook, packt publishing online resources sl.no website link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/iot/homepage 5 http://esp32.net/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with

Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/ □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Raspberry PI
             Outline the features of embedded computing
  M4.01                                                           2
Understanding
             boards - Raspberry PI
             Illustrate the interfacing of basic sensors with
  M4.02                                                           8
Understanding
             embedded computing board - Raspberry PI
             Case Study - the applications building with
             IoT - Smart Perishable tracking/Smart
             transportation, Smart Healthcare, Smart
  M4.03                                                          10
Understanding
             Lavatory maintenance, Smart water through
             IoT, Smart warehouse monitoring, Smart
             Retail, Smart Driver assistance system etc

             Series Test – II                                     1

Contents:
Programming Raspberry Pi with python, Interfacing sensors and actuators with
Raspberry
PI.
Case Study - Applications building with IoT- Smart Perishable tracking/Smart
transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through
IoT,
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Understanding. (IoT). Interpret IoT hardware, protocols and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding. (IoT). Interpret IoT hardware, protocols and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding. software using IoT Stack. Summarize different IoT enabling. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding. software using IoT Stack. Summarize different IoT enabling*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding. technologies. Classify IoT based on the complexity to build. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding. technologies. Classify IoT based on the complexity to build*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding. and operate.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding. and operate.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Module 2

Define or explain Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 3 Understanding boards - Arduino. Illustrate the interfacing of sensors and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding boards - Arduino. Illustrate the interfacing of sensors and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 5 Applying actuators with Arduino Illustrate the interaction

with web page and. (3 marks)

Answer: Begin with the standard definition or identity of *5 Applying actuators with Arduino Illustrate the interaction with web page and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Understanding. Python Illustrate the working of control structures in. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding. Python Illustrate the working of control structures in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Applying. Python Develop python programs to solve simple. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying. Python Develop python programs to solve simple*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Applying. applications. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying. applications*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board -. (3 marks)

Answer: Begin with the standard definition or identity of *Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board -*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart. (3 marks)

Answer: Begin with the standard definition or identity of *8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S

Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/. (3 marks)

Answer: Begin with the standard definition or identity of *10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding. (IoT). Interpret IoT hardware, protocols and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding. Python Illustrate the working of control structures in**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://nptel.ac.in/courses/106/105/106105166/
- https://www.raspberrypi.org/blog/getting-started-with-iot/
- https://learn.adafruit.com/category/internet-of-things-iot https://www.arduino.cc/en/IoT/HomePage http://esp32.net/

IN-PAGE SEARCH

Search this lesson